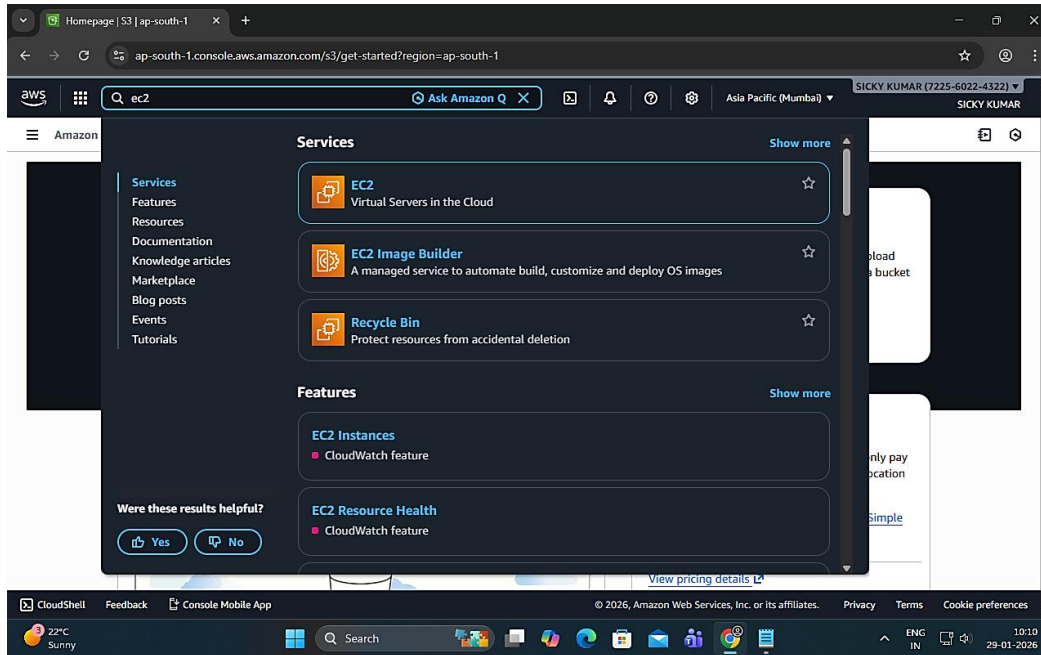


Assignment No: 07**Problem Statement:** Hosting a Website on EC2.**Step 1: Open EC2 Service**

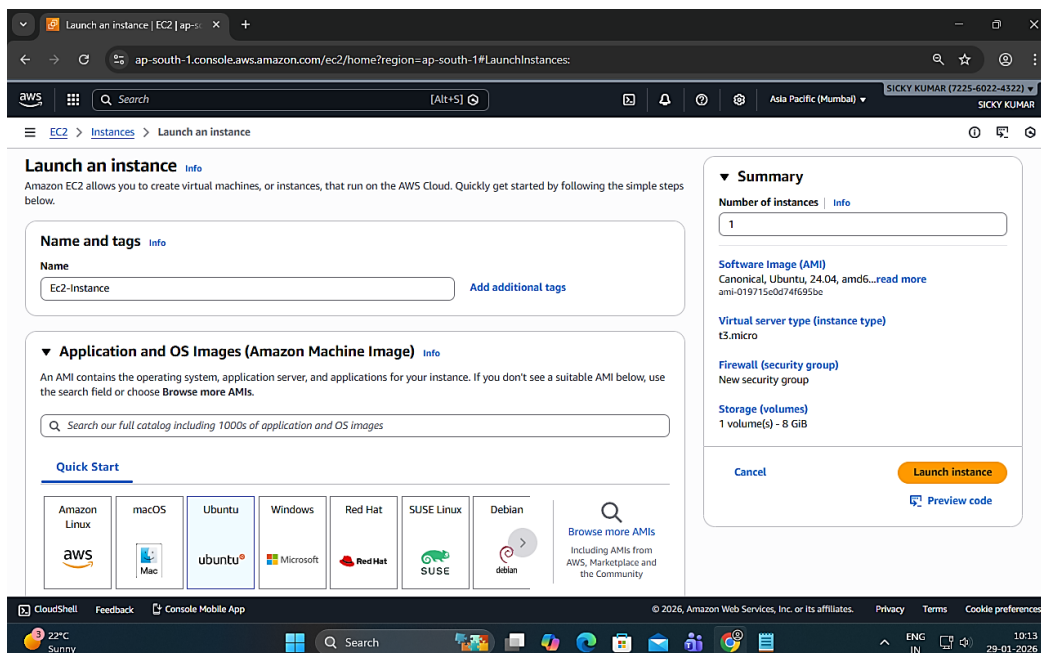
Log in to the AWS Management Console.

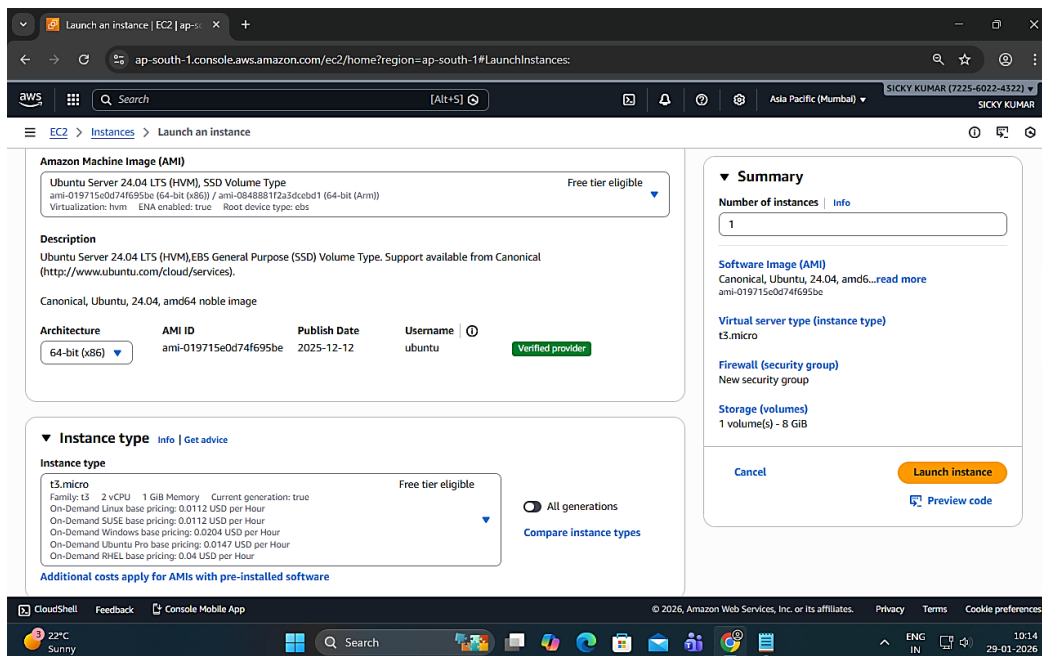
Search for **EC2** in the top search bar and open the **Amazon EC2** service.

This service allows you to create and manage virtual servers.

**Step 2: Launch a New EC2 Instance**Click **Launch Instance**.

Provide a name for the instance (Ec2-Instance).

Choose an **Amazon Machine Image (AMI)** such as **Amazon Ubuntu**.Select an **Instance Type** (t3.micro, free-tier eligible).

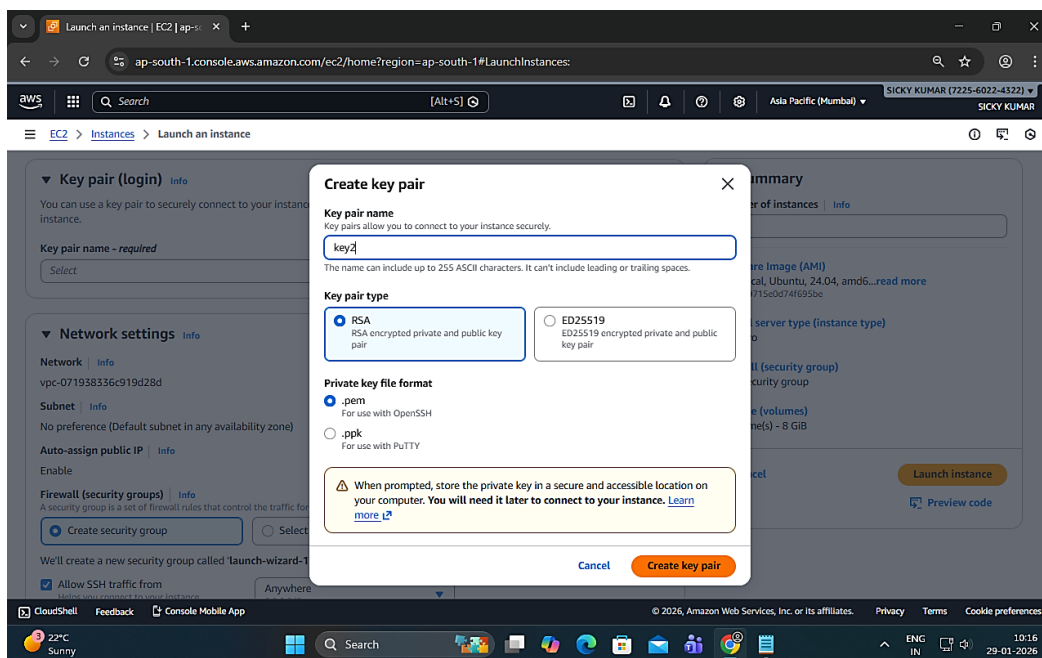


Step 3: Create or Select a Key Pair

Under **Key pair**, select an existing key pair or create a new one.

Download the key pair file (key2.pem) and keep it safe.

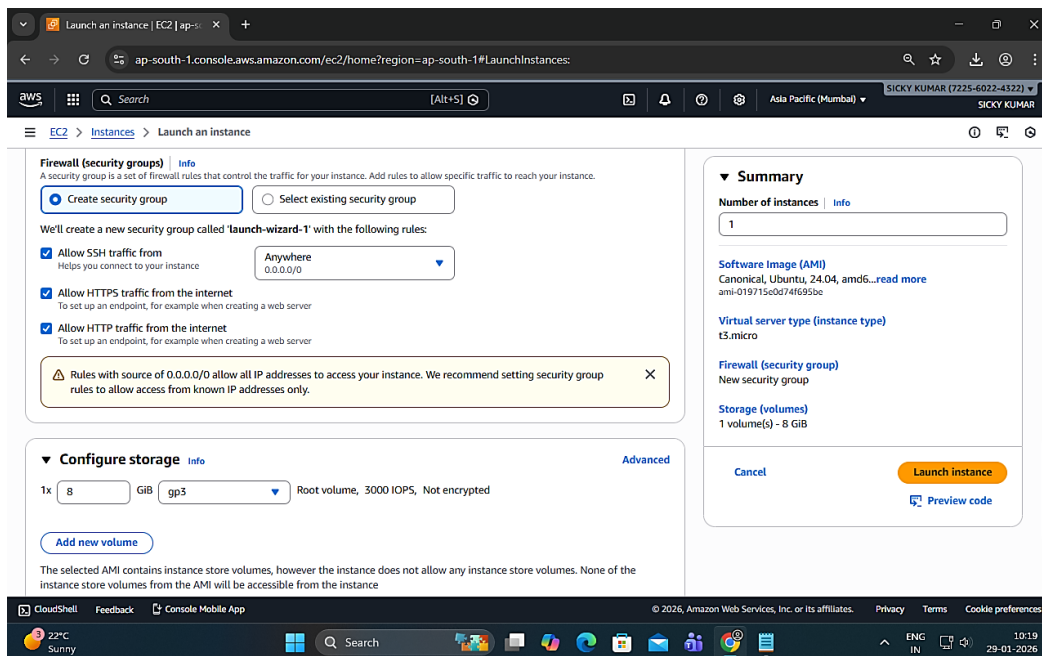
This key is required to connect to the EC2 instance securely.



Step 4: Configure Security Group

Create a new security group and allow the following inbound rules:

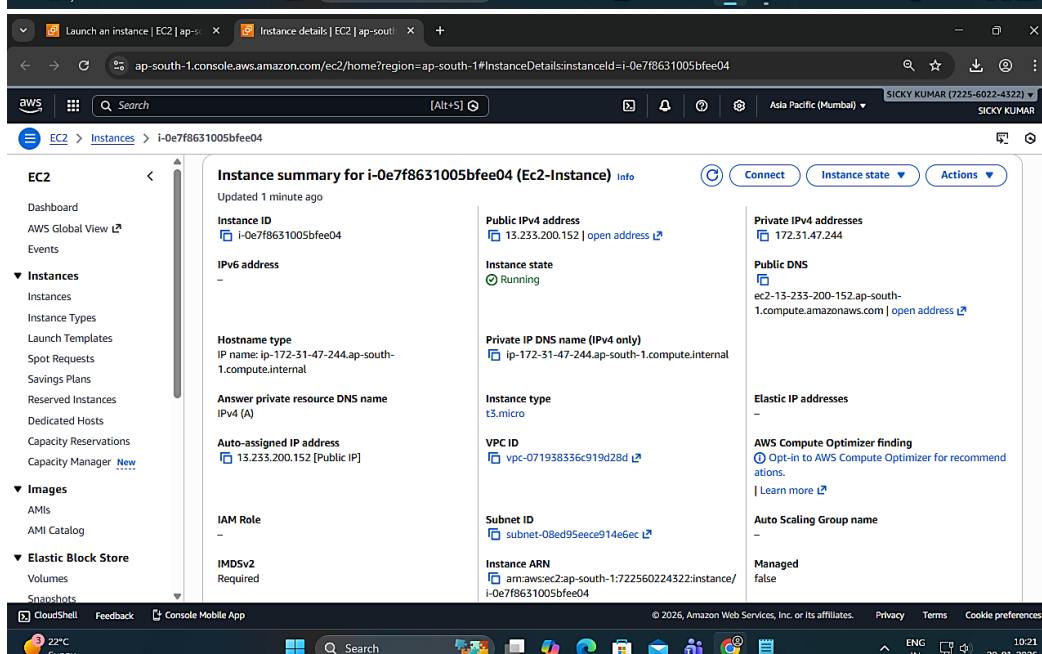
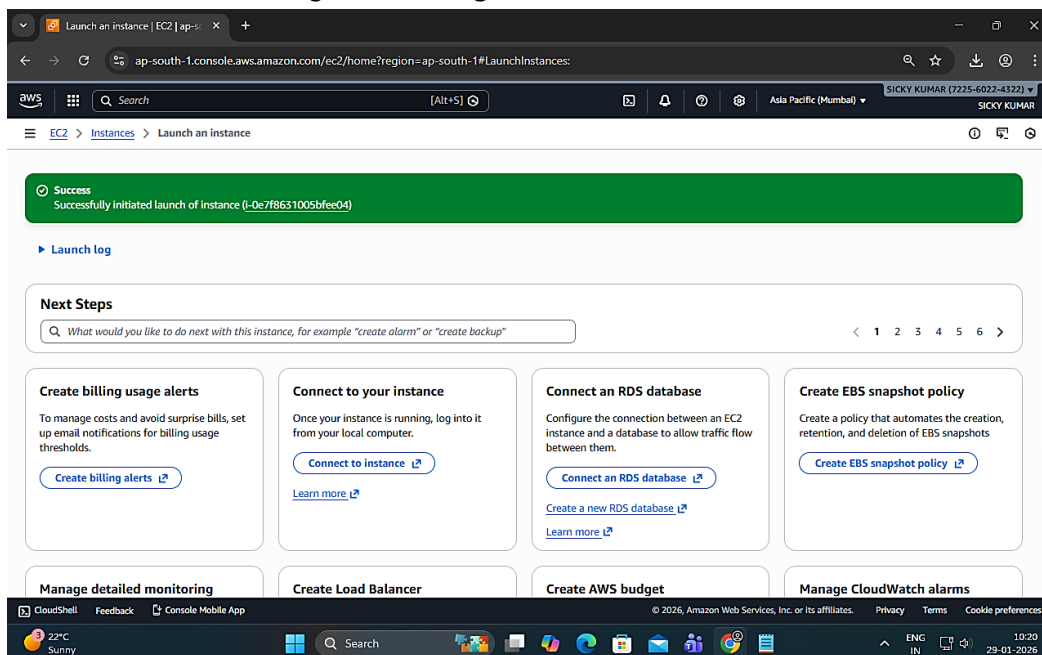
- **SSH (Port 22)** – for remote login
 - **HTTP (Port 80)** – for website access
- Source can be set to **Anywhere (0.0.0.0/0)**.



Step 5: Launch the Instance

Review all settings and click **Launch Instance**.

Wait until the instance state changes to **Running**.



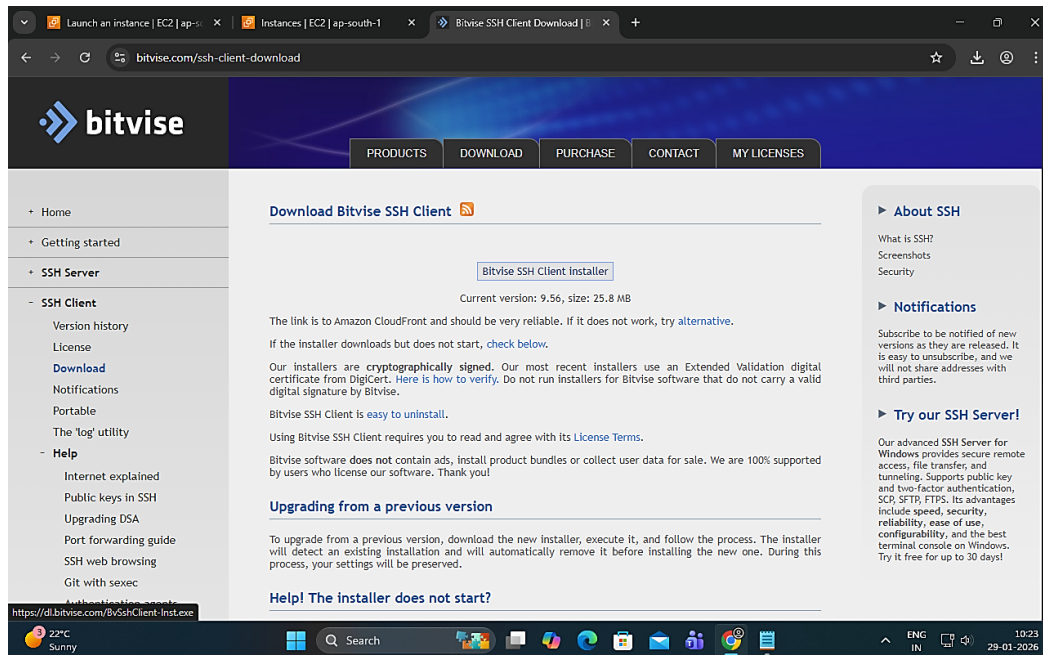
Step 6: Connect to the EC2 Instance Using Bitvise SSH Client (v9.56)

Bitvise SSH Client is a GUI-based tool used to securely connect to an EC2 instance using SSH.

6.1 Download and Install Bitvise

Open a browser and go to the official **Bitvise** website.

Download **Bitvise SSH Client (v9.56)** and install it on your system.

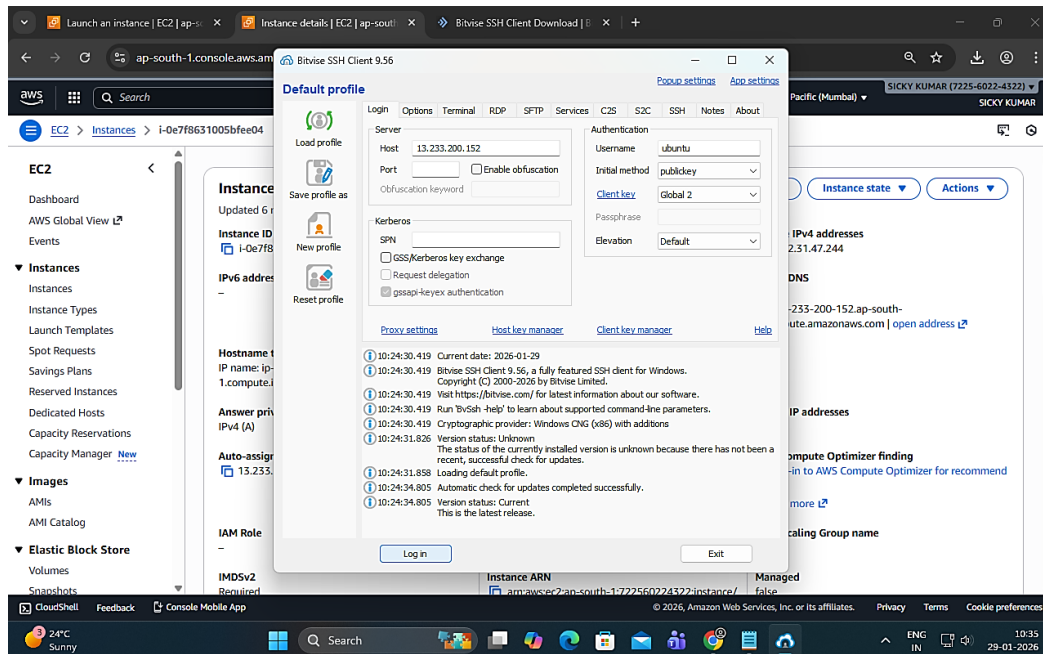


Step 6.2: Configure Bitvise and Load Key Pair

Open Bitvise SSH Client and in the Login tab enter:

- Host: Public IPv4 address (e.g., 13.233.200.152)
- Port: 22
- Username: ubuntu

Under Authentication, select Publickey, click Client key manager, import the .pem key file, click OK, and ensure the key is selected.



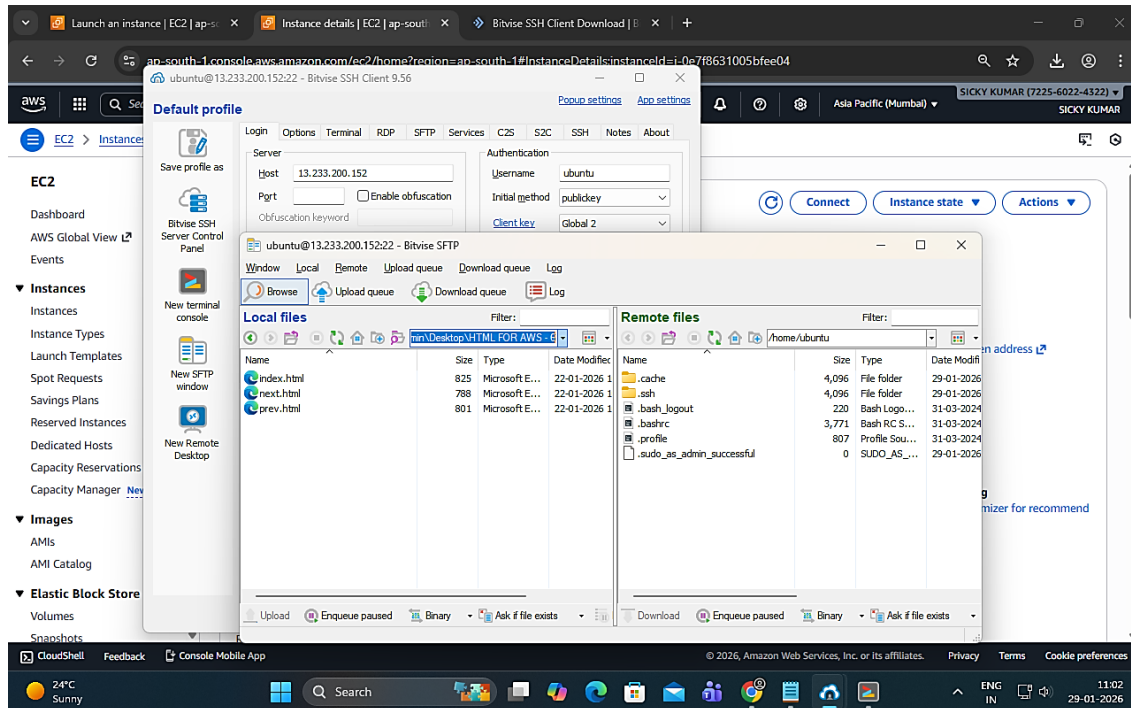
6.3 Connect to EC2

Click **Login**.

Accept the host key when prompted.

The terminal window opens.

✅ EC2 instance is successfully connected using Bitvise SSH Client.



Step 7: Install and Configure Nginx Web Server

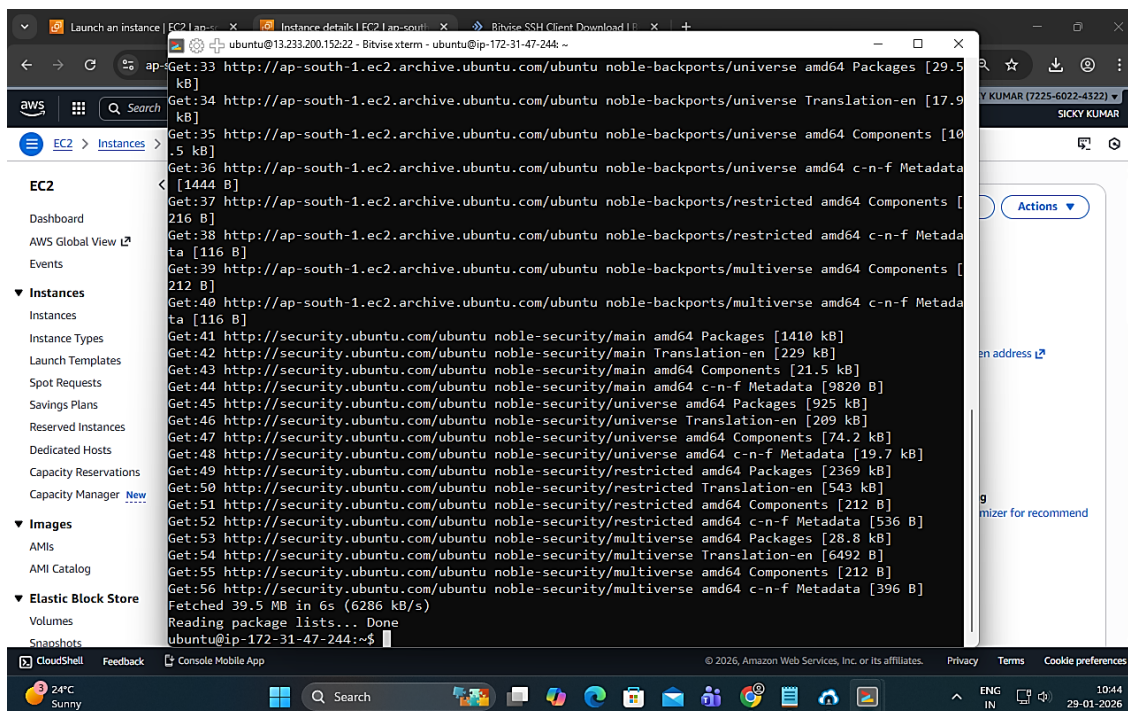
After connecting to the EC2 instance, install and start **Nginx**, which is a lightweight and high-performance web server.

1. Update the system packages:

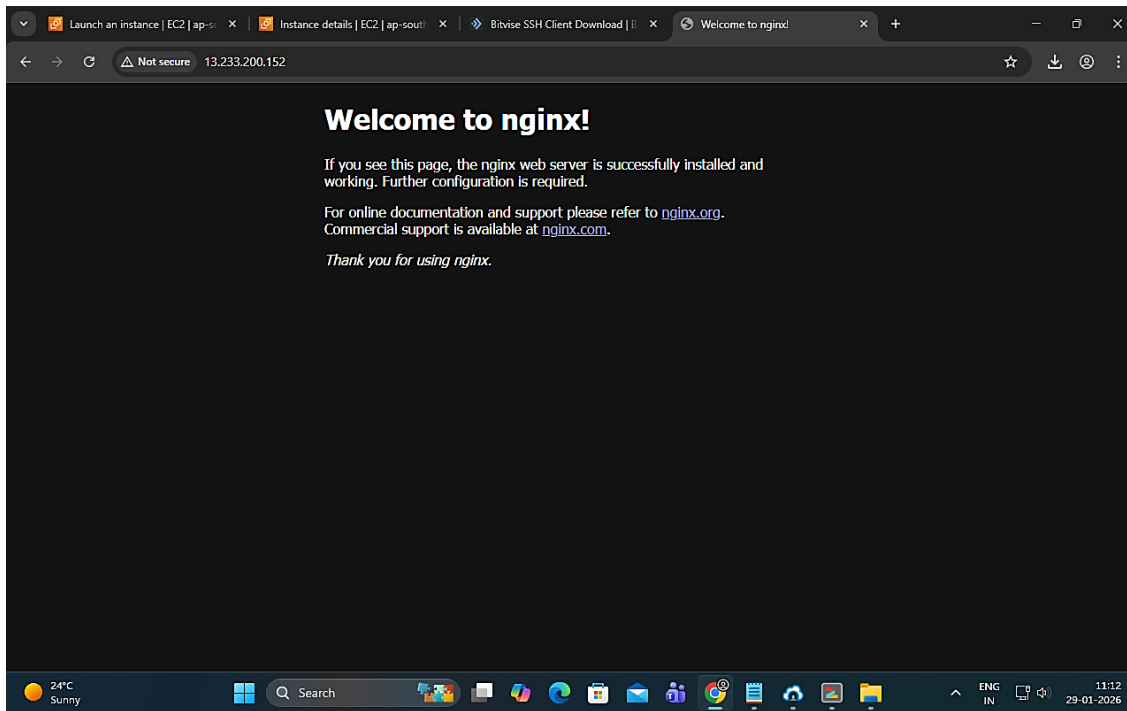
```
sudo apt upgrade -y
```

2. Install Nginx:

```
sudo apt install nginx -y
```



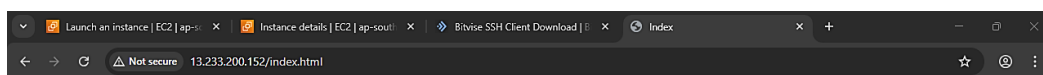
3. Verify Nginx is running by opening the EC2 Public IPv4 address in a browser.



Step 8: Give Directory Permission and Upload Website Files Using Bitvise.

After connecting to the EC2 instance using Bitvise SSH Client:

1. Open the Terminal in Bitvise and give permission to the web directory:
cd /var/www
sudo chmod 777 html
2. Open the SFTP window in Bitvise.
3. On the Remote side, go to:
/var/www/html
4. From the Local side, select three HTML files and drag and drop them into the remote directory.
5. Refresh the browser using the EC2 Public IPv4 address to view the uploaded pages.
6. Permissions are set first then website files are uploaded successfully.



Hello, This is Index Page

